Prerequisites for using (must read)

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1. Overseas version of OpenRouter
    OpenRouter operation steps
2. Overseas version of Qwen

Example cases related to LLM, important premise: need to connect to a wifi network that can access the Internet, how to connect
For Raspberry Pi 4B, please go to "10. Hotspot Mode and Connecting to WiFi" in the Raspberry Pi System Configuration section for specific operations
For Raspberry Pi 5B, please go to "11. Hotspot Mode and Connecting to WiFi (pi5 optional)" in the Raspberry Pi System Configuration section for specific operations

This tutorial is about the configuration of LLM. If you want to run large model related cases normally, you need to follow the steps below to configure it.

1. Overseas version of OpenRouter

Some online large models of the overseas version of the boot example use the middleware of the OpenRouter platform, which is a third-party platform. This large model can be called for free 50 times a day. If you want to continue using it, you can refer to the usage rules for expansion.

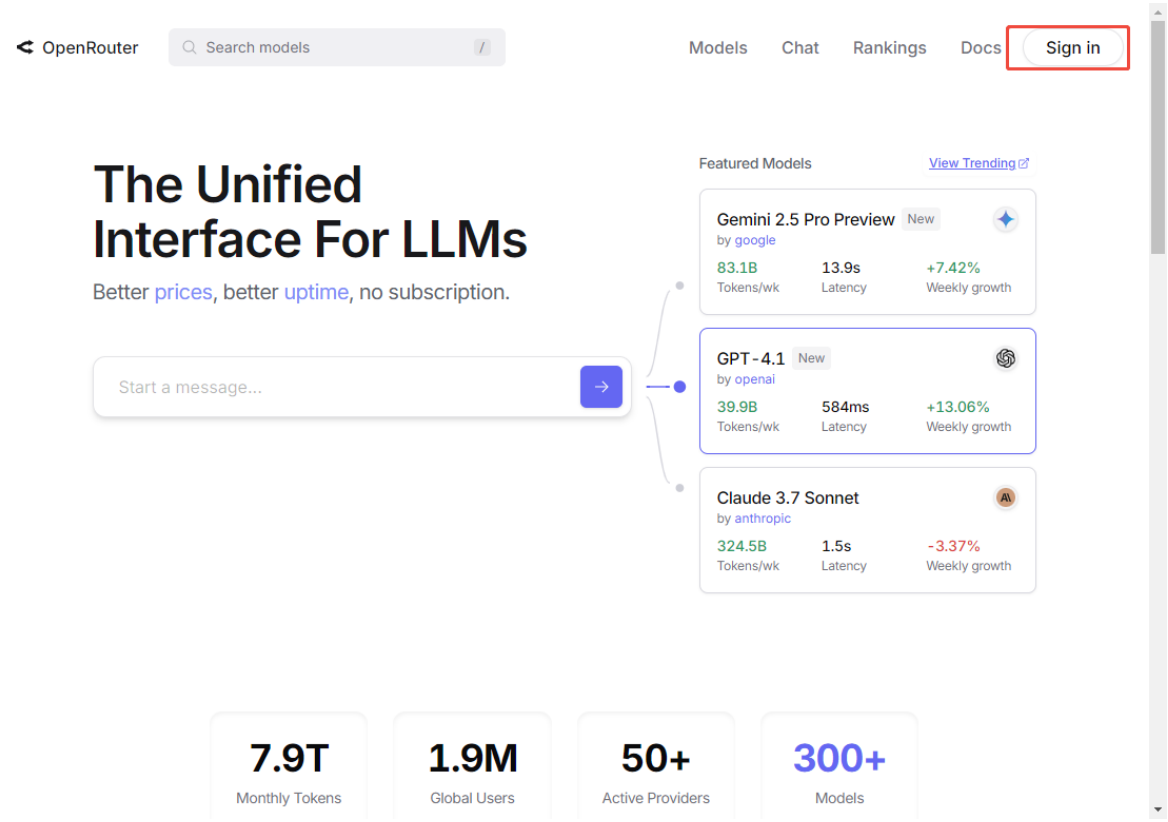
The following cases require the use of this large model: Q&A, Embodied intelligence, Smart dog, and Picture to text

OpenRouter operation steps

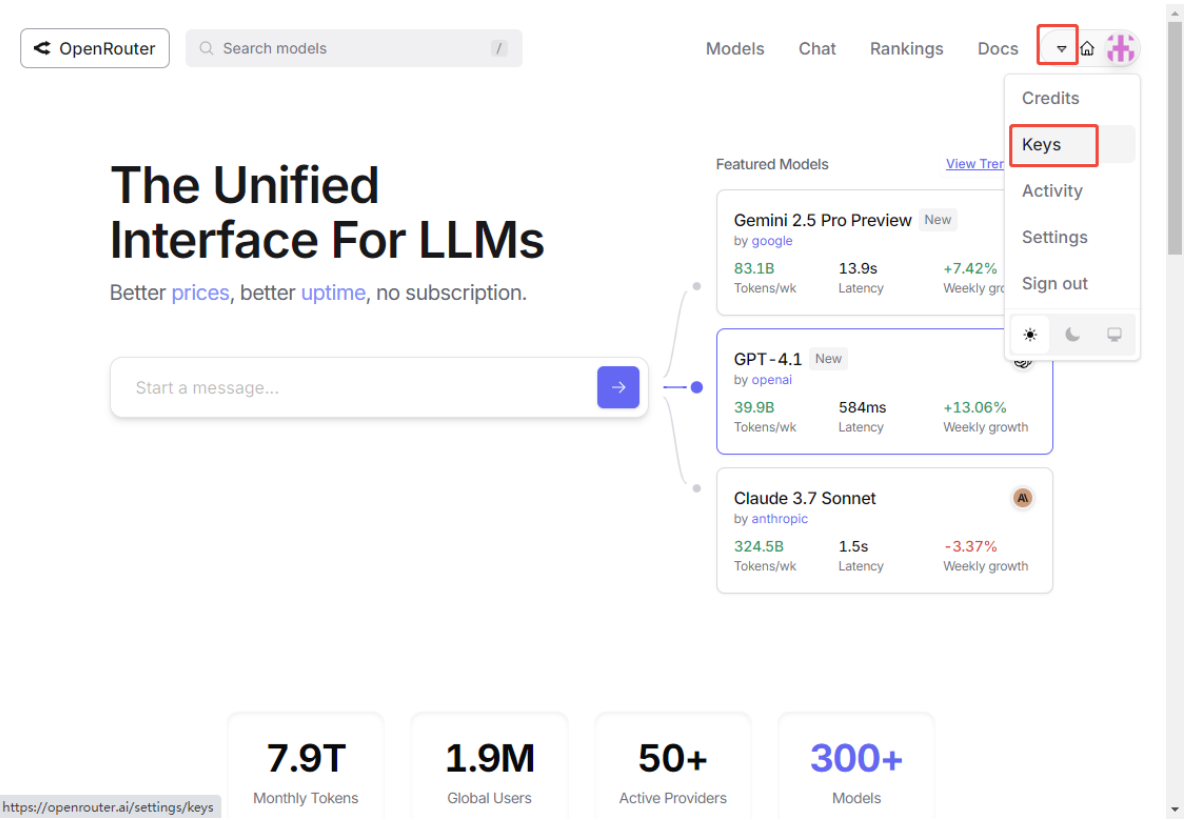
1. First log in to the following official website

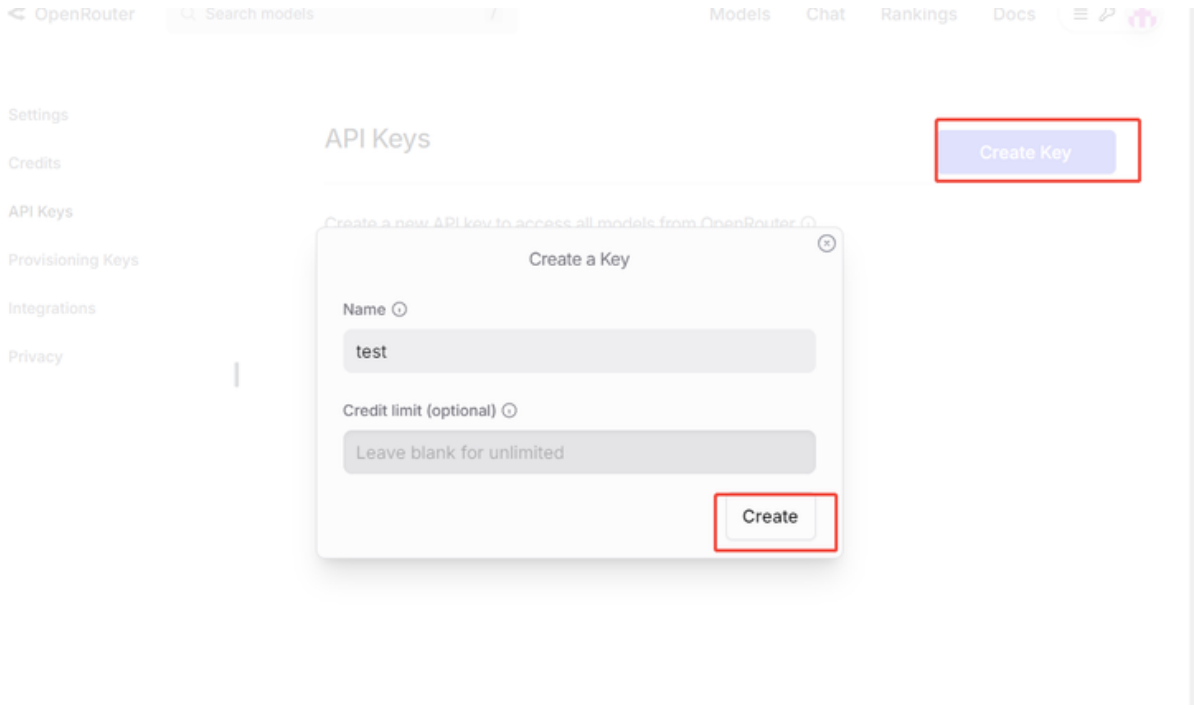
https://openrouter.ai/

2. Then log in. You can use multiple ways to log in. This article recommends using a github account to log in to this platform.

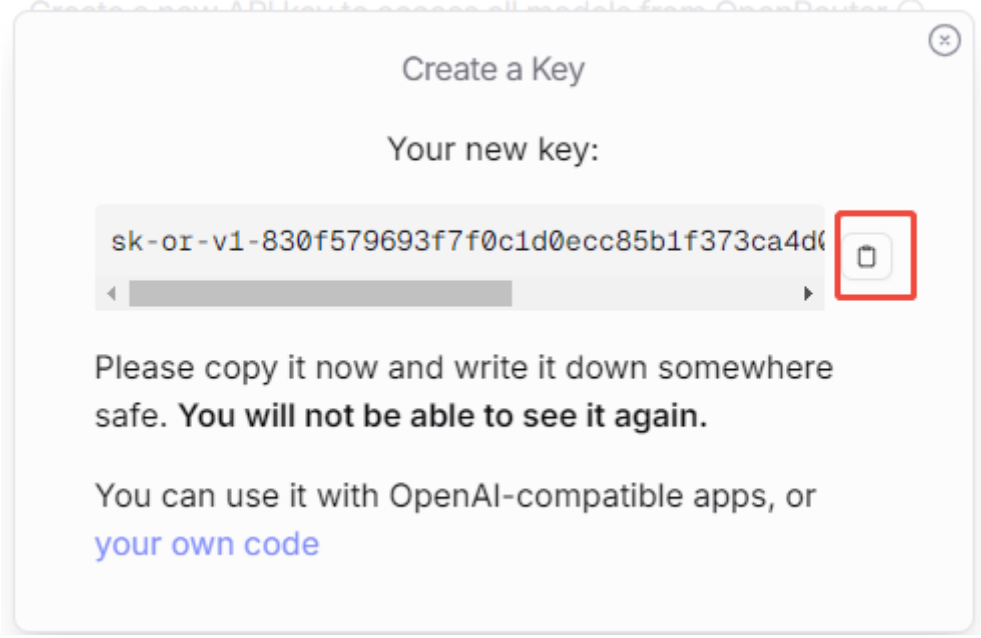


3. Enter the API-KEY page and create one

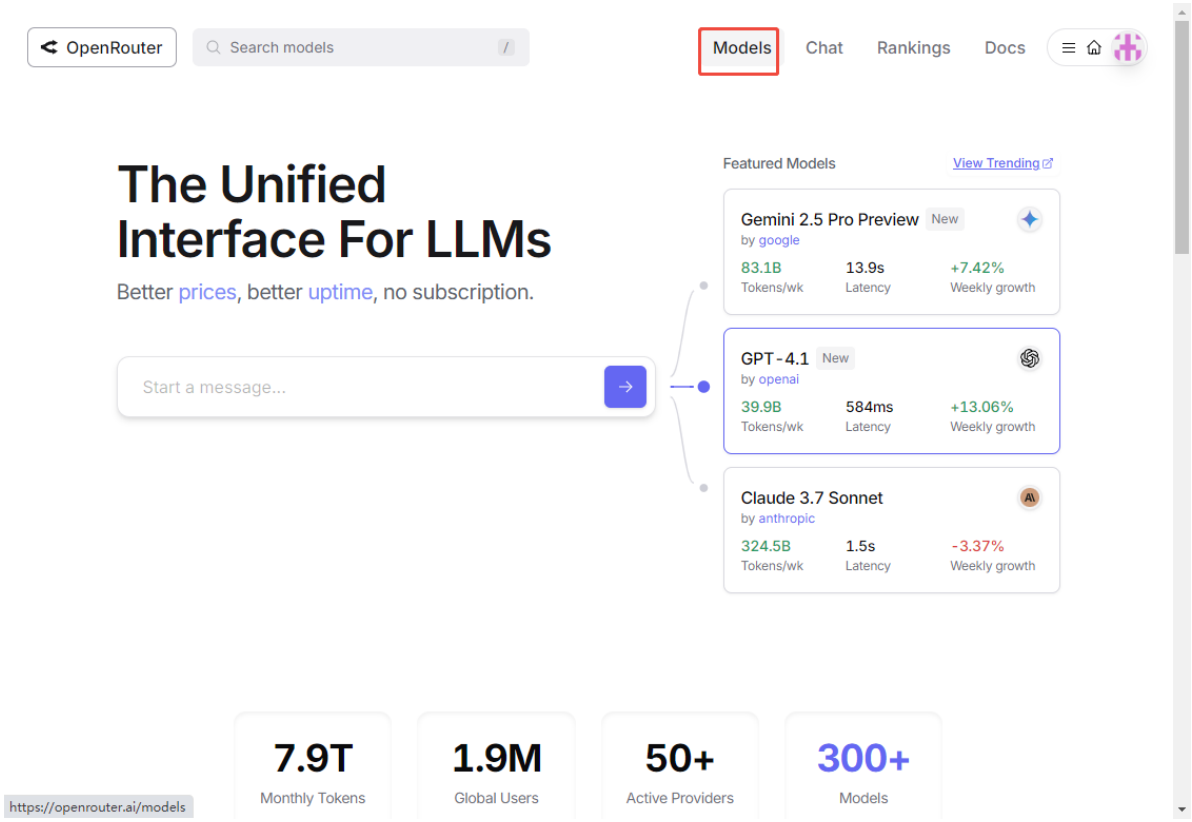


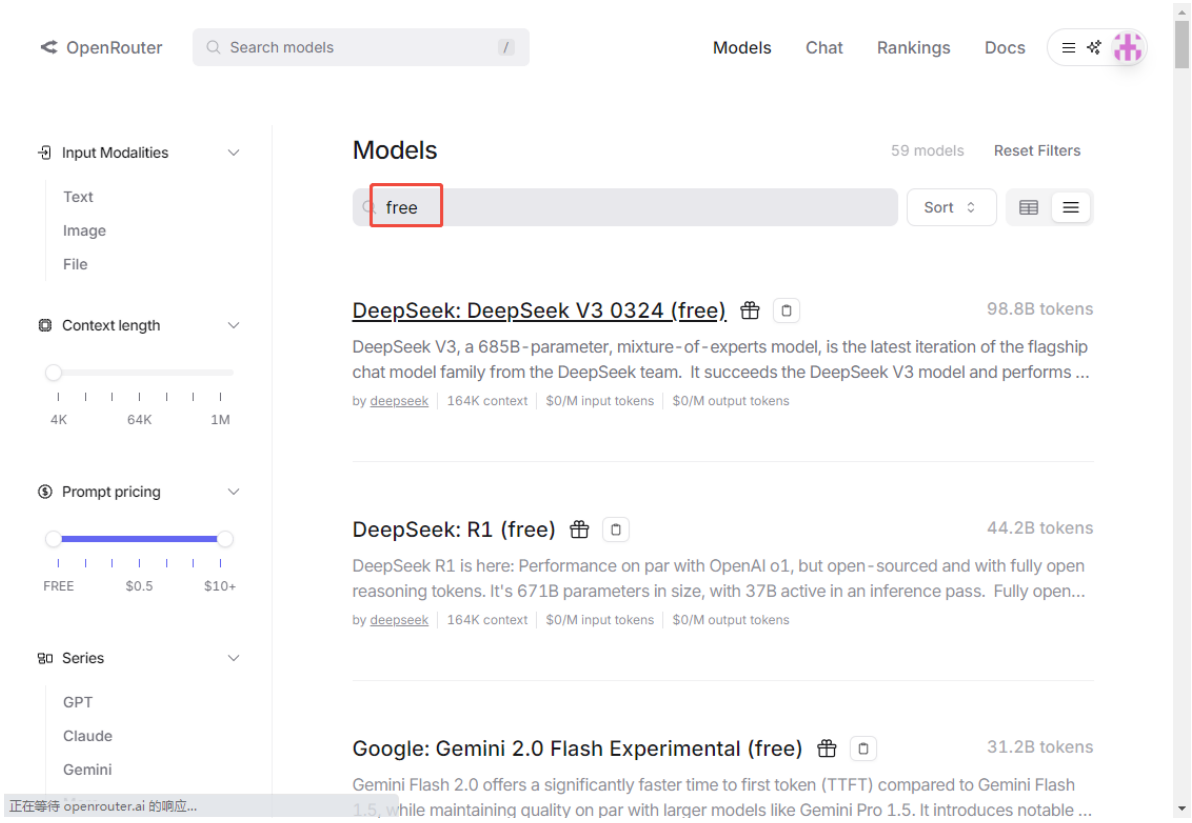


4. Then copy the created key and save it on your computer. **You can only copy this key when you create it for the first time**. You cannot view it again after leaving this interface.

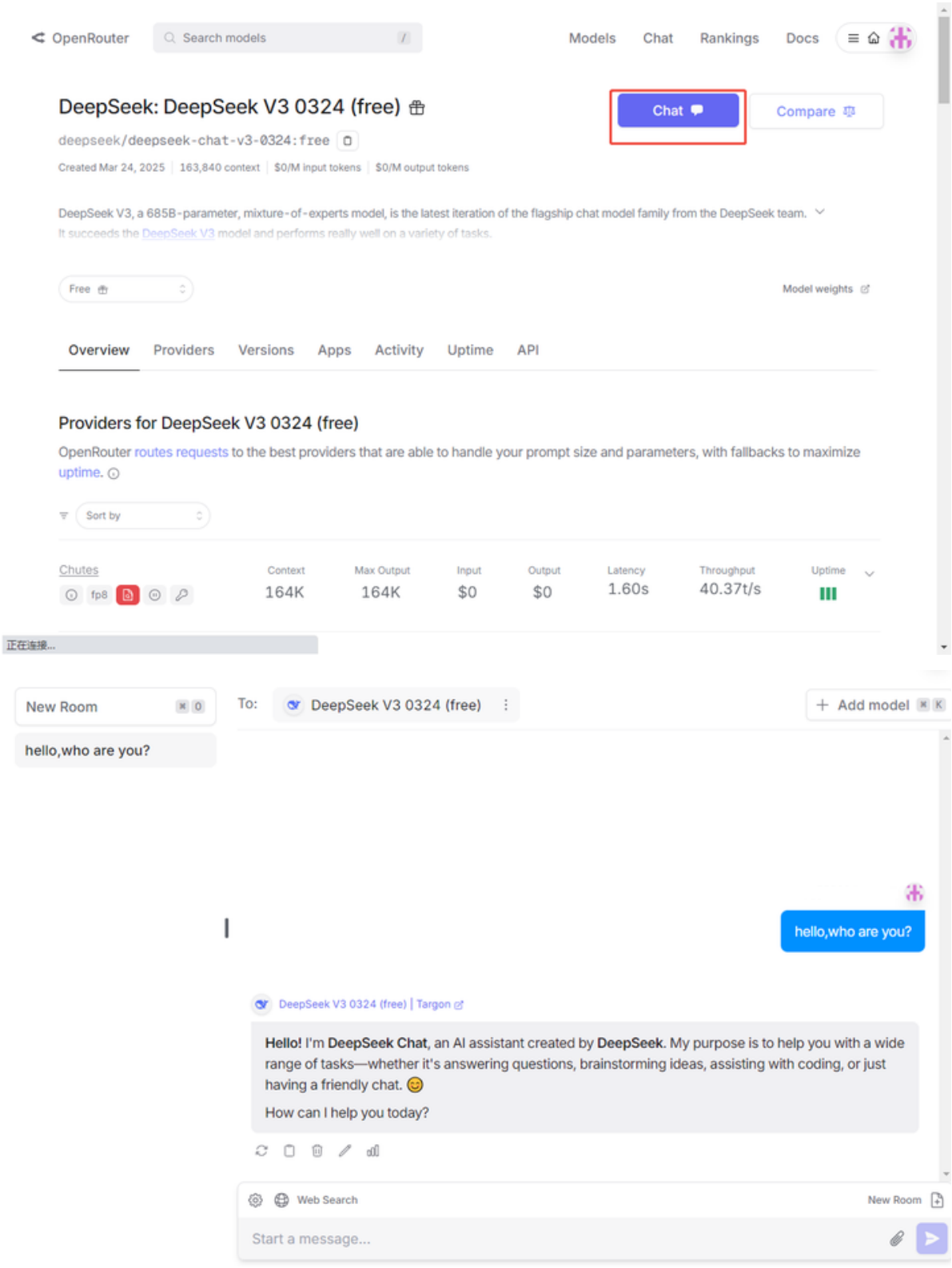


5. Then we find a free model to test whether the registered account can be used normally





6. If it is a normal reply, it means it can be used normally.



7. The created API-KEY can directly use the free model. If you want to use the non-free model, you need to purchase the non-free model with your credit card. There is no tutorial or explanation, so you can study it yourself.

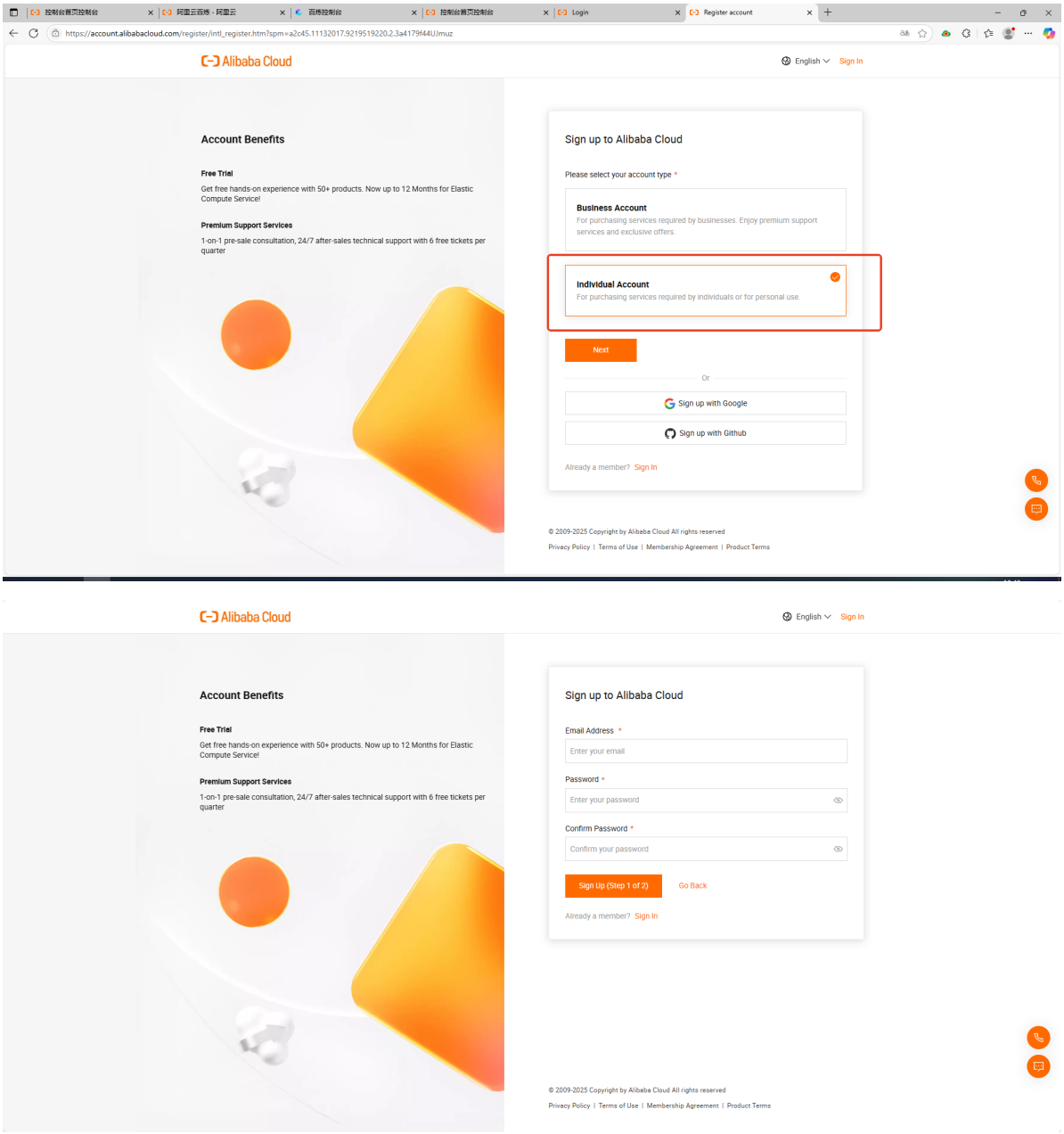
2. Overseas version of Qwen

Some online large models in the overseas version startup example use the visual large model of Qwen. This platform is a third-party platform, and the large model is a powerful visual model provided by Alibaba. After the registration and authentication is completed, some large models can be used for free, and more models can be used according to this website.

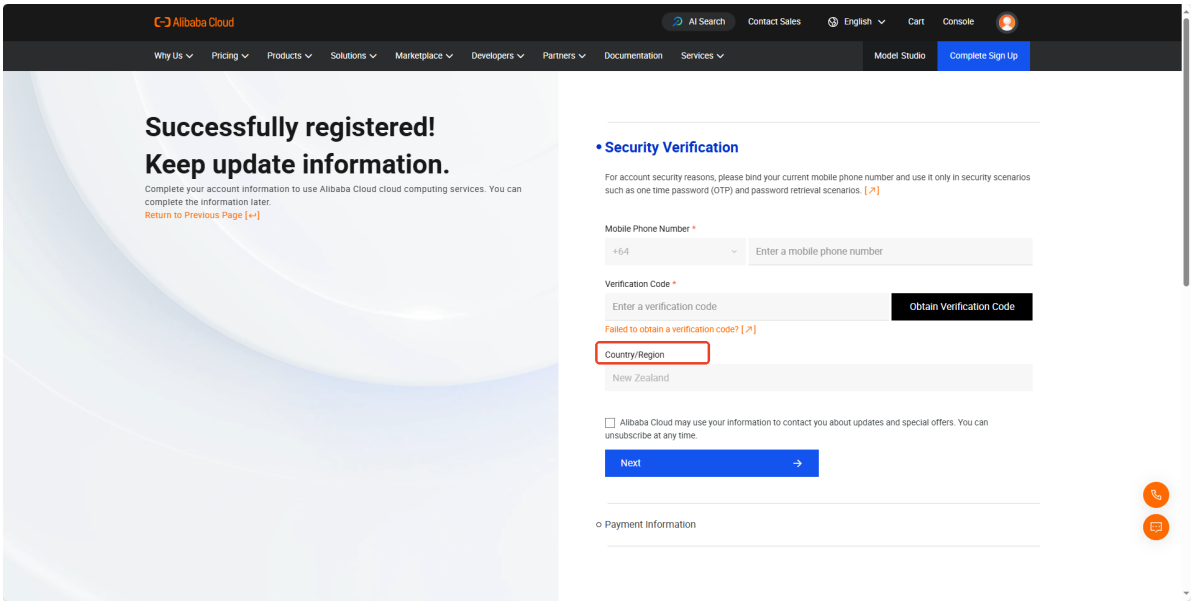
The following cases require the use of this large model: Control object tracking

1. Open the website and register

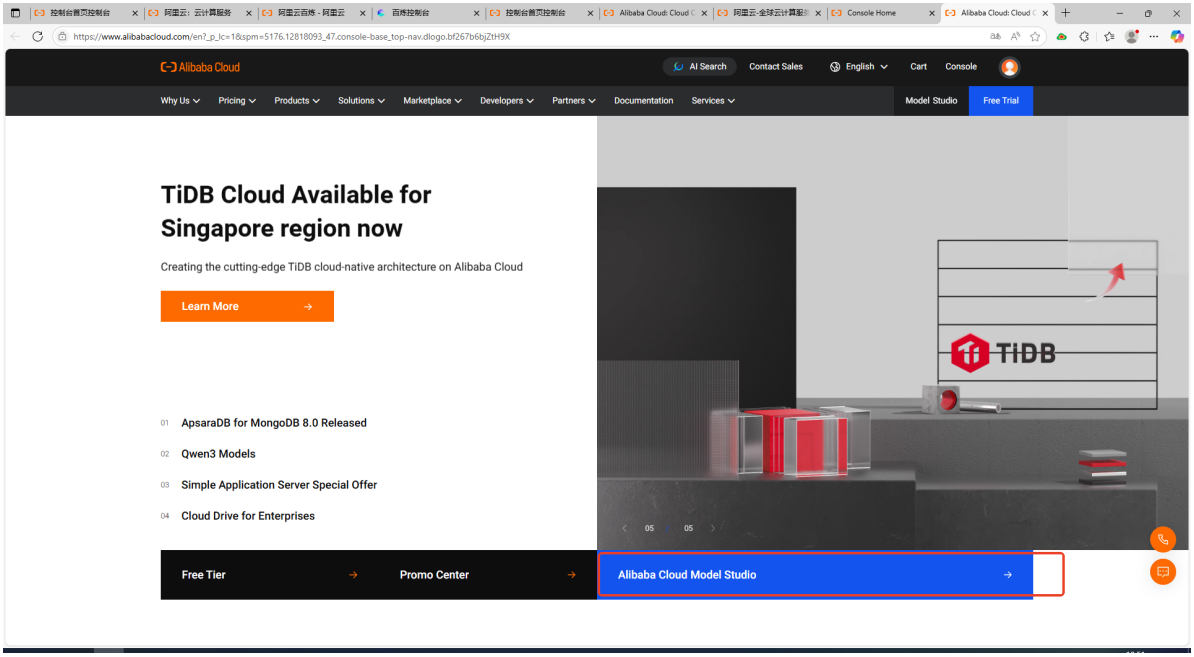
[Register account]

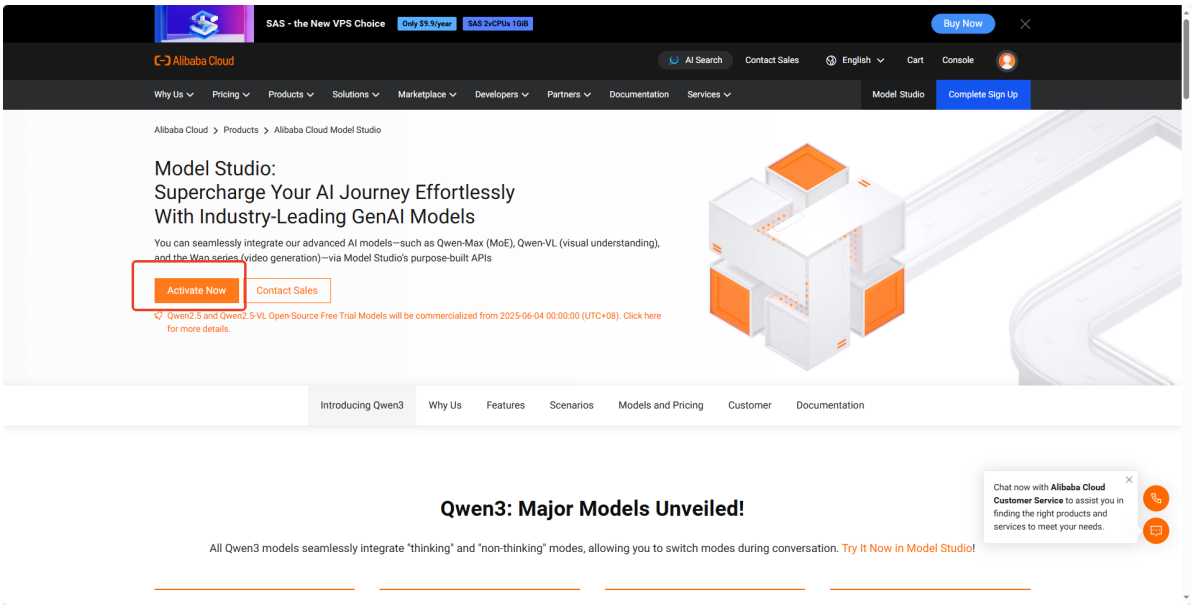


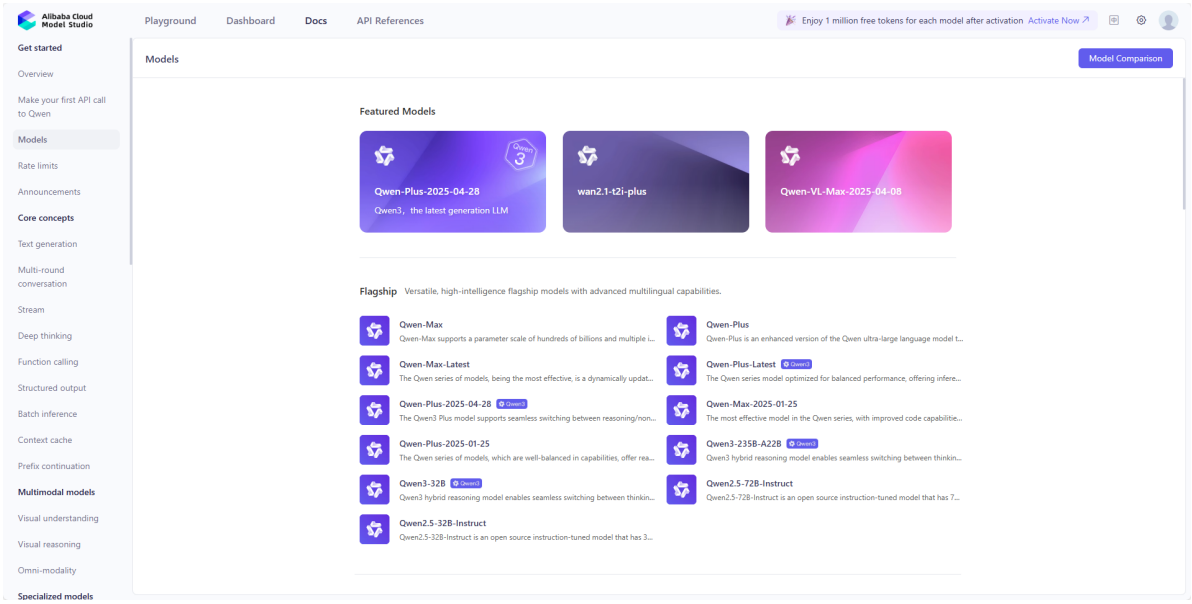
2. After registration, enter the contact information according to your country and get certified. Without certification, you cannot get the free large model

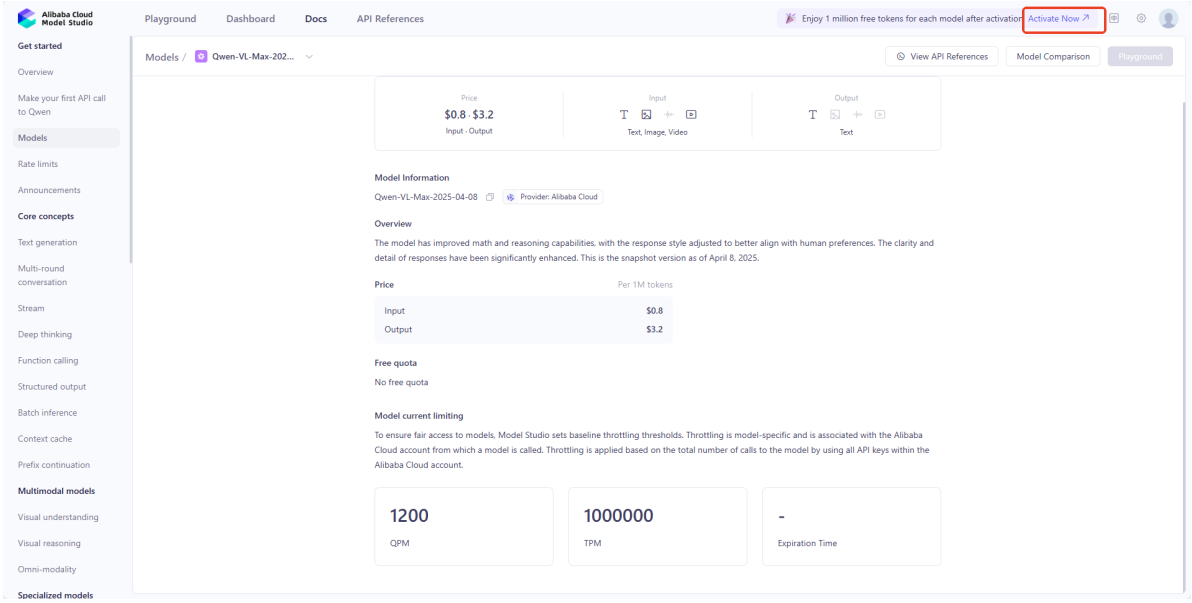


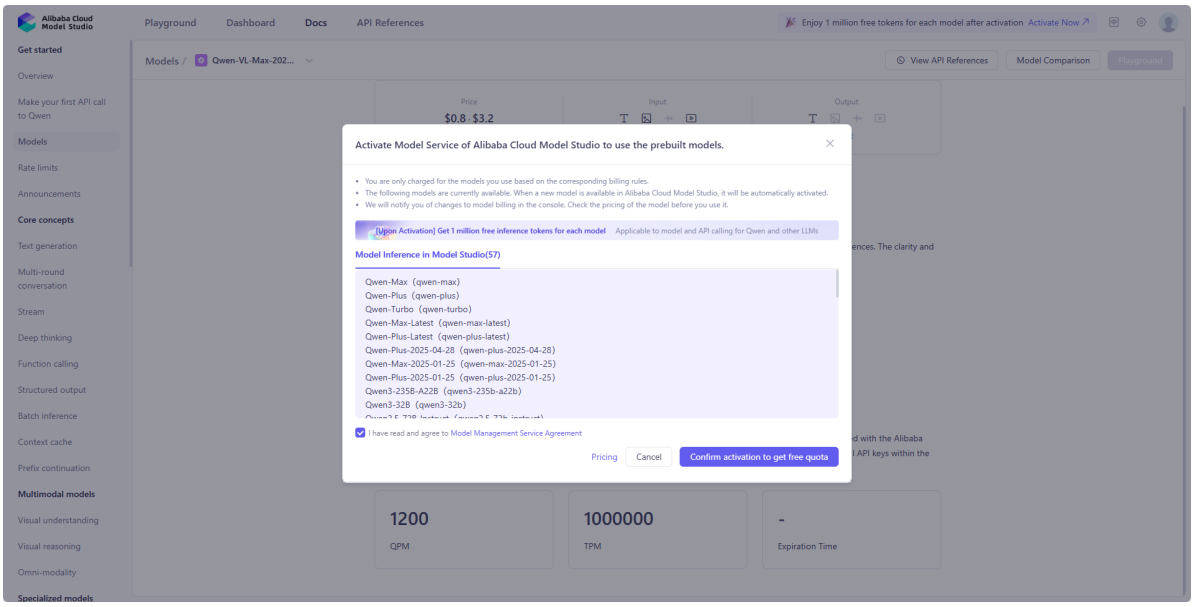
3. Enter the large model, select Qwen-VL-Max, and get it for free Take part of the large model quota



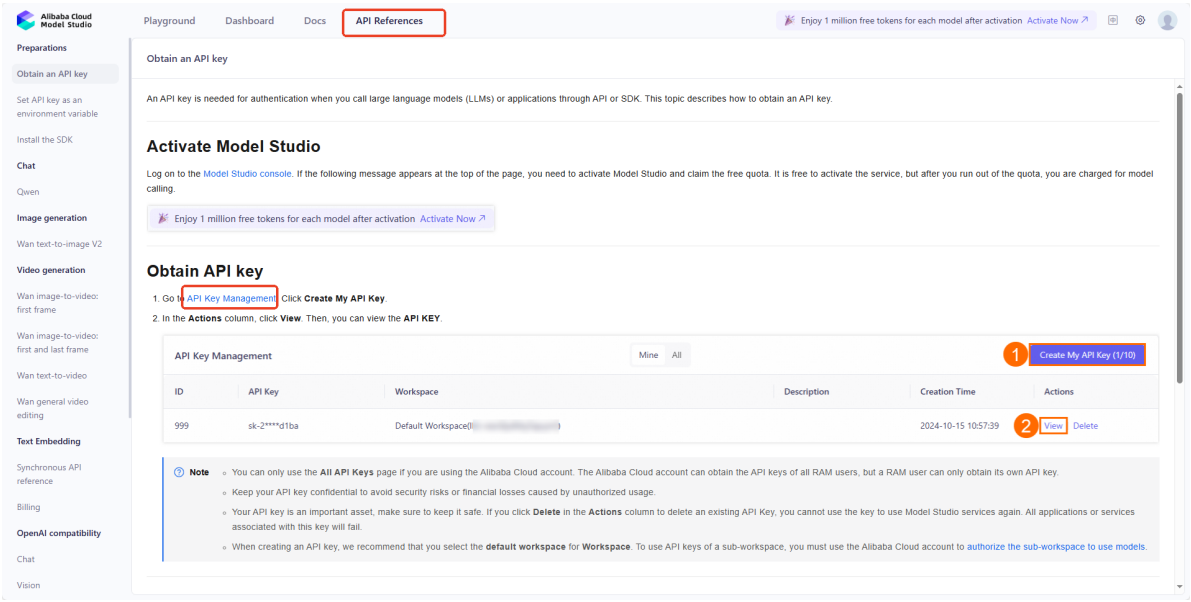


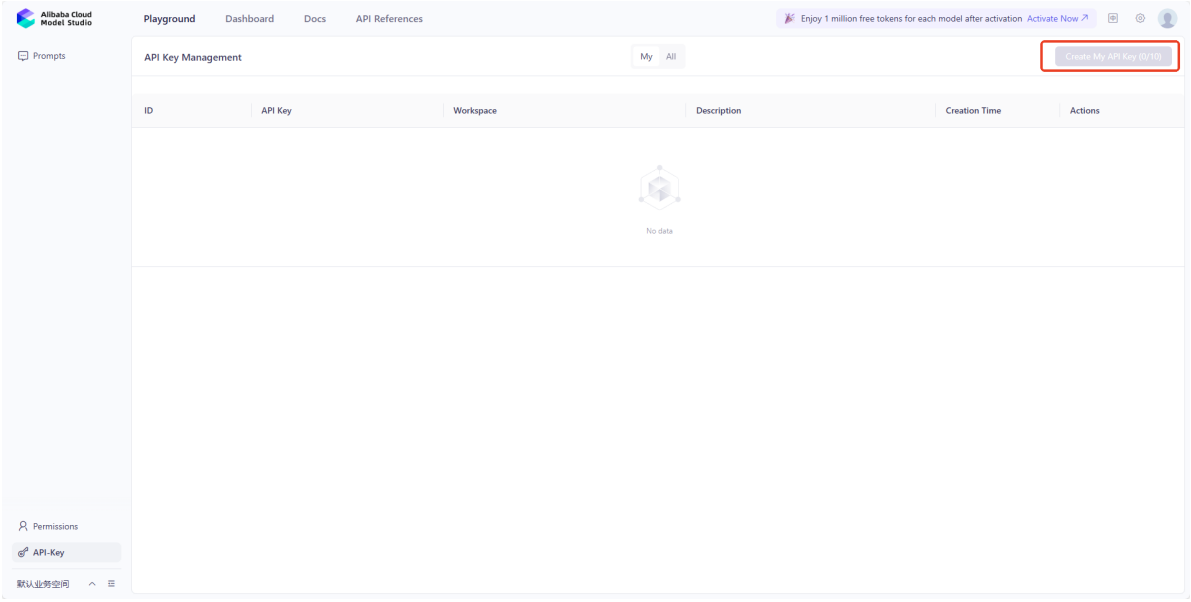






4. Register your own API Key





Then remotely access the robot dog system through vnc and open the terminal. For how to enter the system through VNC, you can watch the tutorial in Chapter 1.

6. Configure API key, enter in terminal

```
nano /home/pi/DOGZILLA/Samples/4_Big_Modle/API_KEY.py
```

```
####国内key相关的 Domestic key related
#星火大模型 Starfire Large Model
XINGHOU_APPID = 'xxxxxx' #填写星火大模型的APPID
XINGHOU_APISecret = 'xxxxxx' #填写星火大模型的APISecret
XINGHOU_KEY = "xxxxxxxx" #填写星火大模型的APIKEY


#通义千问 Tongyi Qianwen
TONYI_key='xxxxxxxxx' #填写通义千问的APIKEY

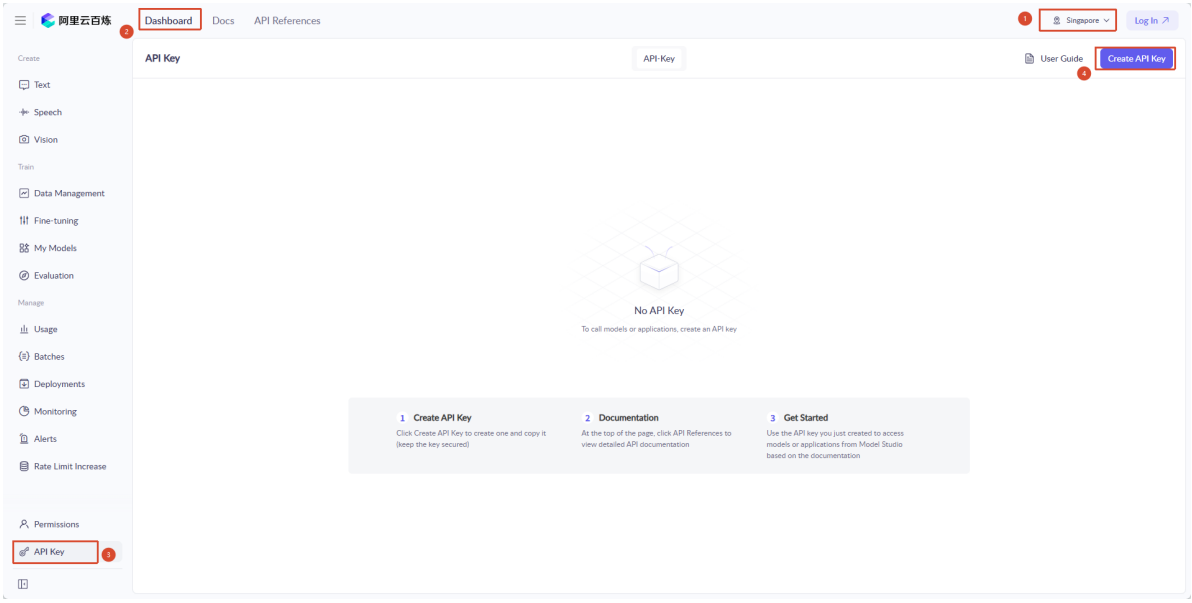


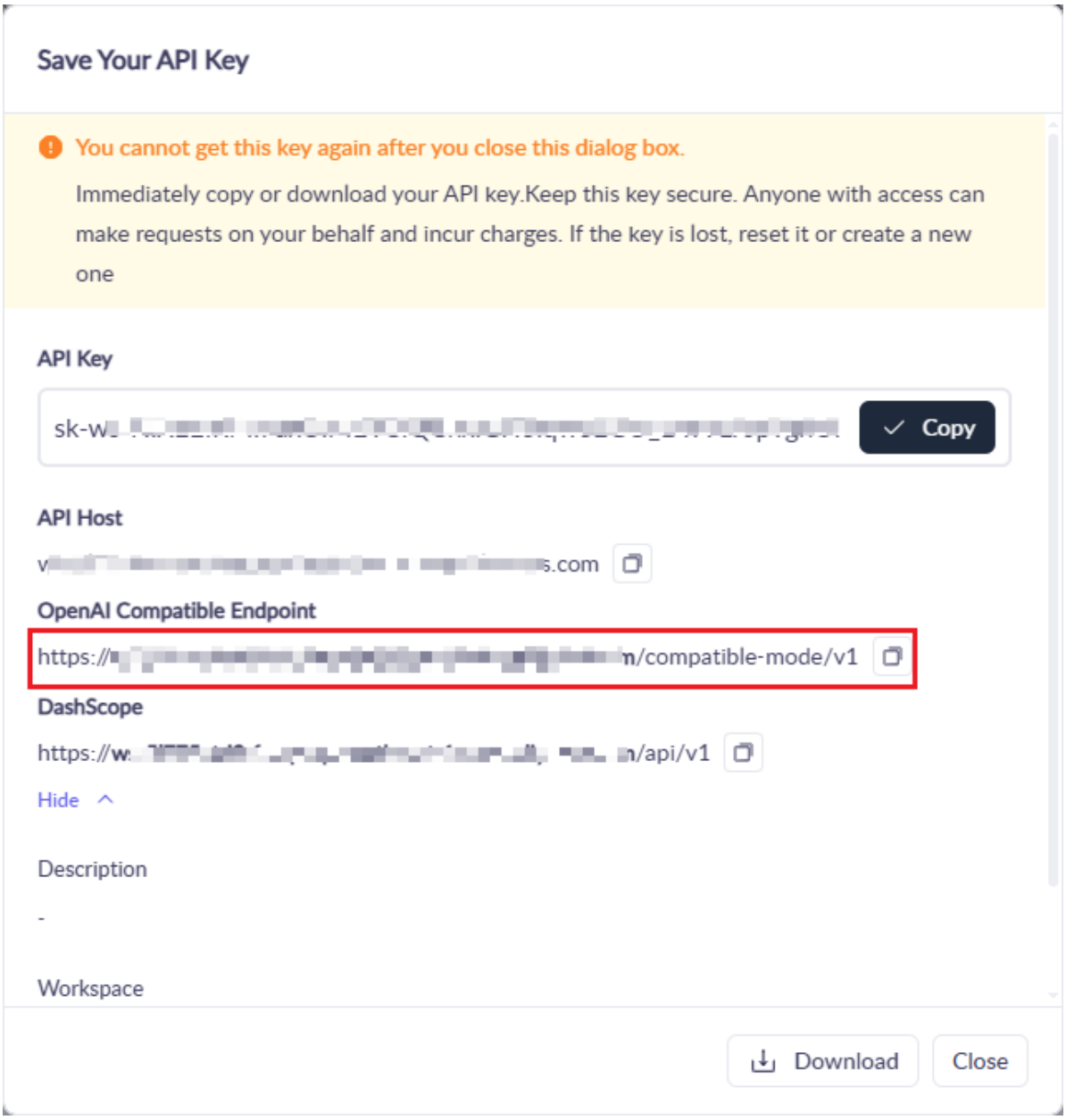
#####国外key相关的  Foreign key related
#注册的地址: https://openrouter.ai/   Registered address: https://openrouter.ai/
openAI_KEY = 'xxxxxxx' #填写openrouter的APIKEY
```

Then enter the API-KEY of OpenRouter and the API-KEY of TONYI_KEY according to the prompts. You do not need to fill in the Spark Big Model. Then press Ctrl+S to save and Ctrl+X to exit.
The Spark Big Model key used by overseas customers has been packaged in the relevant cases and does not need to be filled in here.

7. Configure the base URL for Tongyi Qwen

Users need to configure the `tongyi_base_url` corresponding to their account. Go to the Model Service Platform Bailian Console to apply for an API key. Copy the API key and OpenAI-compatible address (Base URL), then fill them into the corresponding fields.





Enter the following command in the terminal to open the dog_tongyiAPI.py file, then paste the copied OpenAI-compatible address in the base_url field.

```python
#  base 64 编码格式
def encode_image(image_path):
    with open(image_path, "rb") as image_file:
        return base64.b64encode(image_file.read()).decode("utf-8")

def QwenVL_api_picture(PROMPT='机械狗管家智能体'):
    base64_image = encode_image("./dog_agent/rec.jpg")
    client = OpenAI(
        # 若没有配置环境变量，请用百炼API Key将下行替换为：api_key="sk-xxx"
        api_key= TONYI_key,
        base_url="https://dashscope.aliyuncs.com/compatible-mode/v1",
    )
    completion = client.chat.completions.create(
        model="qwen-vl-max-latest",  #qwen-vl-max-latest
        messages=[
            {
                "role": "user",
                "content": [
                    {
                        "type": "image_url",
                        "image_url": {"url": f"{base64_image}"},
                    },
                    {"type": "text",
                     "text": PROMPT
                    },
                ],
```

Enter the following command in the terminal to open the dog_Track_food_api.py file, then paste the copied OpenAI-compatible address in the base_url field.

```
nano nano
/home/pi/DOGZILLA/Samples/4_Big_Modle/ros_dog_agent/dog_Track_food_api.py
```

```python
#  base 64 编码格式
def encode_image(image_path):
    with open(image_path, "rb") as image_file:
        return base64.b64encode(image_file.read()).decode("utf-8")

def QwenVL_api_picture(PROMPT='追踪积木块上方的物体'):
    base64_image = encode_image("./ros_dog_agent/myrec.jpeg")
    client = OpenAI(
        # 若没有配置环境变量，请用百炼API Key将下行替换为：api_key="sk-xxx"
        api_key= TONYI_key,
        base_url="https://dashscope.aliyuncs.com/compatible-mode/v1",
    )
    completion = client.chat.completions.create(
        model="qwen-vl-max-latest",
        messages=[
            {
                "role": "user",
                "content": [
                    {
                        "type": "image_url",
                        "image_url": {"url": f"{base64_image}"},
                    },
                    {"type": "text",
                     "text": SYSTEM_PROMPT + PROMPT
                    },
                ],
            }
        ],
```

Then press Ctrl+S to save and Ctrl+X to exit.

Enter the following command in the terminal to open the dog_Track_food_api.py file, then paste the copied OpenAI-compatible address in the base_url field.

```
nano nano /home/pi/DOGZILLA/Samples/4_Big_Modle/dog_agent/dog_Track_food_api.py
```

```python
#  base 64 编码格式
def encode_image(image_path):
    with open(image_path, "rb") as image_file:
        return base64.b64encode(image_file.read()).decode("utf-8")

def QwenVL_api_picture(PROMPT='追踪积木块上方的物体'):
    base64_image = encode_image("./dog_agent/myrec.jpeg")
    client = OpenAI(
        # 若没有配置环境变量，请用百炼API Key将下行替换为：api_key="sk-xxx"
        api_key= TONYI_key,
        base_url="https://dashscope.aliyuncs.com/compatible-mode/v1",
    )
    completion = client.chat.completions.create(
        model="qwen-vl-max-latest",
        messages=[
            {
                "role": "user",
                "content": [
```

Then press Ctrl+S to save and Ctrl+X to exit.

Enter the following command in the terminal to open the dog_tongyiAPI.py file, then paste the copied OpenAI-compatible address in the base_url field.

```
nano nano /home/pi/DOGZILLA/Samples/4_Big_Modle/dog_agent/dog_tongyiAPI.py
```

```python


#  base 64 编码格式
def encode_image(image_path):
    with open(image_path, "rb") as image_file:
        return base64.b64encode(image_file.read()).decode("utf-8")

def QwenVL_api_picture(PROMPT='机械狗管家智能体'):
    base64_image = encode_image("./dog_agent/rec.jpg")
    client = OpenAI(
        # 若没有配置环境变量，请用百炼API Key将下行替换为：api_key="sk-xxx"
        api_key= TONYI_key,
        base_url="https://dashscope.aliyuncs.com/compatible-mode/v1",
    )
    completion = client.chat.completions.create(
        model="qwen-vl-max-latest",  #qwen-vl-max-latest
        messages=[
            {
                "role": "user",
                "content": [
                    {
                        "type": "image_url",
                        "image_url": {"url": f"{base64_image}"},
                    },
                    {"type": "text",
                     "text": PROMPT
                    },
```

Then press Ctrl+S to save and Ctrl+X to exit. You can run the Big Model routine.